

Assignment – 11:

Problem Statement:

Write a program in C to accept a number from the user and compute the following:

- a) Square root of the number
- b) Square of the number
- c) Cube of the number
- d) Check whether the number is prime
- e) Factorial of the number
- f) Prime factors of the number

Code:

```
#include <stdio.h>

#include <math.h>

int isPrime (int n)
{
    if (n<=1)
    {
        return 0;
    }
    for (int i=2;i<=n/2;i++)
    {
        if (n%i==0)
        {
            return 0;
        }
    }
    return 1;
}

long long factorial (int n)
{

```

```

    long long fact=1;
    for (int i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    return fact;
}

void primeFactors (int n)
{
    printf ("Prime factors: ");
    for (int i=2;i<=n;i++)
    {
        while (n%i==0)
        {
            printf("%d ",i);
            n=n/i;
        }
    }
    printf("\n");
}

int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d",&num);
    printf("Square Root = %.2f\n",sqrt(num));
    printf("Square = %d\n",num*num);
    printf("Cube = %d\n",num*num*num);
    if (isPrime(num))

```

```
{
    printf("Number is Prime\n");
}
else
{
    printf("Number is not Prime\n");
}
if (num>=0)
{
    printf("Factorial = %lld\n",factorial(num));
}
else
{
    printf("Factorial not defined for negative numbers\n");
}
if (num>1)
{
    primeFactors(num);
}
else
{
    printf("No prime factors\n");
}
}
```

Input & Output:

```
Enter a number: 4
Square Root = 2.00
Square = 16
Cube = 64
Number is not Prime
Factorial = 24
Prime factors: 2 2
```

```
Enter a number: 7
Square Root = 2.65
Square = 49
Cube = 343
Number is Prime
Factorial = 5040
Prime factors: 7
```

